Paint-Booth — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager · Quality Worker
> **Applies to:** Line-D, station 4, Paint-Booth

1. Purpose

The Paint-Booth applies the finish topcoat via electrostatic or robotic spray. It feeds the Curing-Oven (Line-D’s bottleneck), so a paint outage quickly back-fills the booth-side buffer and blocks Primer-Application.

2. Normal operating envelope

- Ideal cycle time: **40 s**
- Max acceptable cycle time: **52 s**
- Expected reject rate: **≤ 4 %**
- Fault probability per cycle: **0.8 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Primer-Application (station 3) is not feeding parts.	Verify Primer-Application state.
`Blocked`	Curing-Oven (station 5) cannot accept the painted part.	Verify Curing-Oven state — this is the line bottleneck.

4. Faults & corrective actions

4.1 `Nozzle-Clog`

- **Symptoms in telemetry:** `fault_type = "Nozzle-Clog"`. Pattern asymmetric; coating thickness rejects rise.
- **Likely root cause:** dried paint in the gun tip; supply contamination.

- **Corrective action:**
  1. Purge gun with solvent; clean tip.
  2. Inspect supply filter; replace if loaded.
- **Restart criteria:** pattern matches reference card.

4.2 `Air-Filter-Saturated`

- **Symptoms in telemetry:** `fault_type = "Air-Filter-Saturated"`. Booth airflow drops; overspray accumulates on parts.
- **Likely root cause:** intake or exhaust filter loaded with paint solids past service interval.
- **Corrective action:**
  1. LOTO. Replace saturated filters.
  2. Verify booth pressure differential.
- **Restart criteria:** airflow within spec.

4.3 `Paint-Viscosity-Drift`

- **Symptoms in telemetry:** `fault_type = "Paint-Viscosity-Drift"`. Coating thickness variable; orange peel appearance.
- **Likely root cause:** solvent evaporation in the supply pot, or batch out of spec.
- **Corrective action:**
  1. Sample paint; measure viscosity.
  2. Adjust thinner to recipe.
  3. Re-validate against the test card.
- **Restart criteria:** viscosity within recipe range.

5. Preventive maintenance schedule

- **Daily:** gun purge; viscosity check.
- **Weekly:** filter inspection; booth pressure check.
- **Monthly:** filter changeout.
- **Quarterly:** full atomization-system service.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Safety_Lockout_Tagout_Procedure.pdf`
- `Line-D_03_Primer-Application_SOP.pdf`
- `Line-D_05_Curing-Oven_SOP.pdf`
- `Line-D_06_Coating-Inspection_SOP.pdf`